Exercise 1: Cosmology II - The Friedmann Equations

Recall that in the previous exercise sheet you studied the flat FLRW metric,

$$ds^2 = -dt^2 + a^2(t)\left(dx^2 + dy^2 + dz^2\right) , \tag{1}$$

which you found had

$$R_{tt} = -3\frac{\ddot{a}}{a} , \qquad R_{ii} = 2\dot{a}^2 + a\ddot{a} , \tag{2}$$

with $i = x, y, z$ and $\dot{} = \partial_t$.

a)* We now want to study a Universe filled with a perfect fluid,

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + p g_{\mu\nu} , \tag{3}$$

where U^μ is the fluid's 4-velocity, $U^\mu U_\mu = -1$.

Argue that, if we want to keep homogeneity and isometry, the fluid must be at rest with the FLRW geometry, $U^\mu = (1, 0, 0, 0)$. Show that this implies

$$T^\mu{}_\nu = \mathrm{diag}\left(-\rho, p, p, p\right) . \tag{4}$$

b)* Using the Einstein Equations and the conservation of the stress-energy tensor $\nabla^\mu T_{\mu\nu} = 0$, derive the Friedmann Equations for a Universe filled with a perfect fluid,

$$(F1): \quad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho , \tag{5}$$

$$(F2): \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) , \tag{6}$$

$$(F3): \quad \dot{\rho} = -3\frac{\dot{a}}{a}(\rho + p) . \tag{7}$$

Argue that, if $\rho > 0$ and $p \geq 0$, there cannot be a static solution to the Friedmann Equations.

If the fluid were not at rest, there would be a preferred directio

isotropy.

$$T_{\mu\nu} = (\rho+p)U_\mu U_\nu + p g_{\mu\nu}$$
$$= (\rho+p)\,\mathrm{diag}(1,0,0,0) + p\,\mathrm{diag}(-1,0,0,0)$$
$$= \mathrm{diag}(\rho, p, p, p)$$

Then raising one index yields

$$T^\mu{}_\nu = \mathrm{diag}(-\rho, p, p, p)$$

Einstein Equations (with cosmological constant)

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

EFE $\quad R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = 8\pi G T_{\mu\nu}$

Since R, g and T are all diagonal, the only equations are diagonal

$$-3\frac{\ddot{a}}{a} - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right](-1) = 8\pi G\rho$$

$$\left(2\dot{a}^2 + a\ddot{a}\right) - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right]a^2 = 8\pi G p$$

Conservation law

$$\nabla^\mu T_{\mu\nu} = \nabla^\nu T_{\nu\nu} \text{ (no sum here)}$$
$$= 0$$

$$\nabla^\mu T_{\mu\nu} = \nabla_\mu T^\mu{}_\nu$$
$$= \partial_\mu T^\mu{}_\nu + \Gamma^\mu_{\mu\sigma}T^\sigma{}_\nu - \Gamma^\sigma_{\mu\nu}T^\mu{}_\sigma$$

Recall the only nonzero Christoffels are Γ^0_{ii} and Γ^i_{i0}. Thus for $\nu = 0$

$$\nabla^\mu T_{\mu 0} = \dot{\rho} + 3\frac{\dot{a}}{a}p + 3\frac{\dot{a}}{a}\rho = 0$$

For $\nu = 1, 2, 3$, all terms in the equation vanish. In summary,

$$-3\frac{\ddot{a}}{a} - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right](-1) = -3\frac{\ddot{a}}{a} + \frac{3\ddot{a}}{a} + 3\frac{\dot{a}^2}{a^2}$$
$$= 3\frac{\dot{a}^2}{a^2} = 8\pi G\rho$$
$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G\rho}{3}$$

$$\left(2\dot{a}^2 + a\ddot{a}\right) - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right]a^2 = \left(2\dot{a}^2 + a\ddot{a}\right) - 3\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)a^2$$
$$= 2\dot{a}^2 + a\ddot{a} - 3a\ddot{a} - 3\dot{a}^2$$
$$= -\dot{a}^2 - 2a\ddot{a}$$
$$= 8\pi G p a^2$$

$$\cancel{8}^{4}\pi G p = -\left[\frac{\dot{a}^2}{a^2} + 2\frac{\ddot{a}}{a}\right]$$
$$= -\left[\frac{\cancel{8}^{4}\pi G\rho}{3} + \cancel{2}\frac{\ddot{a}}{a}\right]$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho + 3p\right]$$

$$\dot\rho = -3\frac{\dot a}{a}(\rho+p)$$

c)* Assuming that the fluid has a constant equation of state

$$w = p/\rho\,, \tag{8}$$

solve the Friedmann Equations, that is, find $\rho(a)$ and $a(t)$.

$$\frac{\dot a^2}{a^2} = \frac{8\pi G\rho}{3}$$

$$\frac{\ddot a}{a} = -\frac{4\pi G}{3}\left[\rho+3p\right]$$

$$\dot\rho = -3\frac{\dot a}{a}(\rho+p)$$

First we eliminate p in favour of ρ

$$\frac{\dot a^2}{a^2} = \frac{8\pi G\rho}{3}$$

$$\frac{\ddot a}{a} = -\frac{4\pi G\rho}{3}(1+3w)$$

$$\dot\rho = -3\frac{\dot a}{a}\rho(1+w)$$

Using the second equation to eliminate the first

$$\frac{\ddot a}{\cancel{a}} = -\frac{1}{2}\frac{\dot a^2}{a^{\cancel{2}}}(1+3w)$$

$$\ddot a = \dot a\frac{d\dot a}{da}$$

$$\dot a\frac{d\dot a}{da} = -\frac{1}{2}\frac{\dot a^2}{a}(1+3w)$$

$$\int\frac{1}{\dot a}d\dot a = -\frac{1}{2}(1+3w)\int\frac{1}{a}da$$

$$\ln\dot a = -\frac{1}{2}(1+3w)\ln a + C$$

$$\dot a = Aa^{-\frac{1}{2}(1+3w)}\,,\quad A>0$$

Integrating,

$$a^{\frac{1}{2}(1+3w)}da = A\,dt$$

$$a^{\frac{3}{2}(1+w)} = At + B\,,\quad w\neq-1$$

By redefining $t=0$, we can remove B to find

$$a = A't^{\frac{2}{3}\frac{1}{1+w}}$$

$$\dot\rho = -3\frac{\dot a}{a}\rho(1+w)$$

$$= -3\frac{\dot a}{a}\rho(1+w)$$

$$\frac{d}{dt}\ln\rho = -3(1+w)\frac{d}{dt}\ln a$$

$$\ln\rho = -3(1+w)\ln a + D$$

$$\rho = Ba^{-3(1+w)}$$

d)* Analyze your solution for the three most common values for w: $w=0$ for pressure-less matter, such as dust or cold dark matter (CDM); $w=1/3$ for radiation, such as photons and ultra-relativistic particles like neutrinos; and $w=-1$ for dark energy[1] (this can also be modelled by adding a cosmological constant Λ to the Einstein Equations).

What happens to $a(t)$ when $w=-1$?

Do these three kinds of fluids fail to satisfy any energy condition?

$$a = A't^{\frac{2}{3}\frac{1}{1+w}}\,,\quad w\neq-1$$

$$\rho = Ba^{-3(1+w)}$$

$w=0$: $a = A't^{\frac{2}{3}}$

$$\rho = Ba^{-3}$$

$$\propto t^{-2}$$

Expansion decelerating

$w=\frac{1}{3}$: $a = A't^{1/2}$

$$\rho = Ba^{-4}$$

$$\propto t^{-2}$$

$$w=-1: \quad a^{\frac{1}{2}(1+3w)}da = A\,dt$$

$$a^{-1}\,da = A\,dt$$

$$\ln a = At + B$$

$$a = Fe^{At}$$

$$\rho = \text{const.}$$

Recall for the energy conditions

NEC $\implies (\rho + p) \geq 0$,
WEC $\implies \rho \geq 0$ and $(\rho + p) \geq 0$,
DEC $\implies \rho \geq |p|$,
SEC $\implies (\rho + 3p) \geq 0$ and $(\rho + p) \geq 0$.

$$p = w\rho$$

For $w=-1$ the SEC is violated, $w=0$ and $w=\frac{1}{3}$ do not violate any energy conditions.

Exercise 2: Cosmology III - Slow-roll Inflation

Numerous astronomical observations suggest that the Universe underwent a phase of accelerated ($\ddot{a} > 0$) expansion ($\dot{a} > 0$), called *inflation*, at its birth.

a)* Show that inflation is equivalent to requiring

$$-\dot{H}/H^2 < 1\,, \tag{9}$$

where $H = \dot{a}/a$ is the Hubble parameter.
Argue also that inflation can be modelled using a perfect fluid with

$$w < -1/3\,. \tag{10}$$

What energy condition do we need to violate achieve this?

$$\dot{H} = \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2}$$

$$\frac{\dot{H}}{H^2} = \left(\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2}\right)\left(\frac{a^2}{\dot{a}^2}\right)$$

$$= \frac{\ddot{a}a}{\dot{a}^2} - 1$$

$$-\frac{\dot{H}}{H^2} < 1 \Leftrightarrow -\frac{\ddot{a}a}{\dot{a}^2} < 0$$

Since $a>0$, this is accelerating iff $\ddot{a} > 0$.

We recall from previously that $a^{\frac{3}{2}(1+w)} \propto t$

If $w < -\frac{1}{3}$, then $\frac{3}{2}(1+w) < 1$, thus $\ddot{a} > 0$ and the universe is accelerating.

This violates the strong energy condition

b)* We now model inflation using a scalar field ϕ (the *inflaton*) with potential $V(\phi)$. We still assume homogeneity and isotropy, so the inflaton field can only be a function of time $\phi = \phi(t)$.
Identify the scalar field's energy density ρ_ϕ and pressure p_ϕ

$$\rho_\phi := -T^0_0\,, \tag{11}$$

$$p_\phi \delta^i_j := T^i_j\,, \tag{12}$$

from your results for the $T_{\mu\nu}$ of a scalar field from the previous exercise sheet.
Then, inserting these into the first two Friedmann equations, show

$$(F1): \qquad H^2 = \frac{8\pi G}{3}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right)\,, \tag{13}$$

$$(F2): \qquad \dot{H} = -4\pi G\dot{\phi}^2\,. \tag{14}$$

Finally prove that the Klein-Gordon equation $\Box\phi - V'(\phi) = 0$ yields

$$\ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} = -V'(\phi)\,. \tag{15}$$

We recall that for a minimally coupled scalar field

$$T_{\mu\nu} = \nabla_\mu\phi\nabla_\nu\phi - \frac{1}{2}g_{\mu\nu}\nabla^\rho\phi\nabla_\rho\phi - g_{\mu\nu}V(\phi)\,.$$

$$T^\mu{}_\nu = g^{\mu\sigma}\nabla_\sigma\phi\nabla_\nu\phi - \frac{1}{2}g^{\mu\sigma}g_{\sigma\nu}\nabla^\rho\phi\nabla_\rho\phi - g^{\mu\sigma}g_{\sigma\nu}V(\phi)$$

$$= g^{\mu\sigma}\nabla_\sigma\phi\nabla_\nu\phi - \frac{1}{2}\delta^\mu{}_\nu\nabla^\rho\phi\nabla_\rho\phi - \delta^\mu{}_\nu V(\phi)$$

$$T^0{}_0 = -\dot{\phi}^2 - \frac{1}{2}(-\dot{\phi}^2) - V(\phi)$$

$$= -\frac{1}{2}\dot{\phi}^2 - V(\phi)$$

Hence $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$

Substituting directly yields $H^2 = \frac{8\pi G}{3}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right)$

$$T^1{}_1 = \frac{1}{2}\dot{\phi}^2 - V(\phi)$$

$$\dot{H} = \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2}$$

$$= -\frac{4\pi G}{3}(\rho + 3p) - H^2$$

$$= -\frac{4\pi G}{3}\rho - 4\pi G p - \frac{8\pi G}{3}\rho$$

$$= -4\pi G\rho - 4\pi G p$$

$$= -4\pi G\left[\frac{1}{2}\dot{\phi}^2 + V(\phi) + \frac{1}{2}\dot{\phi}^2 - V(\phi)\right]$$

$$= -4\pi G\dot{\phi}^2$$

Klein-Gordon equation $\nabla_\mu\nabla^\mu\phi - V'(\phi) = 0$

$$\nabla_\mu\nabla^\mu\phi = \nabla_\mu(\partial^\mu\phi)$$

$$= \partial_\mu\partial^\mu\phi + \Gamma^\mu_{\mu\rho}\partial^\rho\phi$$

$$= -\ddot{\phi} - 3\frac{\dot{a}}{a}\dot{\phi}$$

Hence $\quad \ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} = -V'(\phi)$

Exercise 3: Cosmology IV - Einstein's Biggest Blunder?

e) The FLRW metric can be generalized to

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right] = -dt^2 + a^2(t)\left[d\chi^2 + f_k^2(\chi) d\Omega^2\right] , \tag{16}$$

where the parameter k describes the curvature of the spatial slices, and

$$f_k(\chi) = \begin{cases} \sin\chi & \text{for} \quad k = 1 , \\ \chi & \text{for} \quad k = 0 , \\ \sinh\chi & \text{for} \quad k = -1 . \end{cases} \tag{17}$$

If $k > 0$, the Universe is positively curved (a closed 3-sphere), if $k = 0$ the Universe is flat, and if $k < 0$ the Universe is negatively curved (an open 3-hyperboloid).

Moreover, we can add a cosmological constant Λ to the Einstein Equations. Then, the new Friedmann Equations for a Universe filled with a perfect fluid read

$$(F1): \quad \left(\frac{\dot a}{a}\right)^2 = \frac{\Lambda}{3} + \frac{8\pi G}{3}\rho - \frac{k}{a^2} , \tag{18}$$

$$(F2): \quad \frac{\ddot a}{a} = \frac{\Lambda}{3} - \frac{4\pi G}{3}(\rho + 3p) , \tag{19}$$

$$(F3): \quad \dot\rho = -3\frac{\dot a}{a}(\rho + p) . \tag{20}$$

Feel free to rederive these results using `diffgeo` as an extra exercise.

Now, assuming that the Universe is filled only with pressure-less matter ($w = 0$), show that there exists a *static* solution ($\ddot a = 0$ and $\dot a = 0$) to the Friedmann Equations, with a critical density of exactly

$$\rho_c = \frac{\Lambda}{4\pi G} . \tag{21}$$

What must be the curvature of this Universe?

If $\dot a = \ddot a = 0$, and $w=0$, then $p=0$ and the equations reduce to

$$0 = \frac{\Lambda}{3} + \frac{8\pi G}{3}\rho - \frac{k}{a^2}$$

$$0 = \frac{\Lambda}{3} - \frac{4\pi G}{3}\rho$$

$$\dot\rho = 0$$

We simply solve the second equation for

$$\rho_c = \frac{\Lambda}{4\pi G}$$

The first equation yields

$$0 = \frac{\Lambda}{3} + \frac{8\pi G}{3}\frac{\Lambda}{4\pi G} - \frac{k}{a^2}$$

$$= \Lambda - \frac{k}{a^2}$$

Since $a^2 > 0$, we must have $k > 0$ for a solution.

f) Finally, perturb the previous solution $\rho = \rho_c + \delta\rho$, with $\delta\rho \ll \rho_c$, to show that this solution in fact unstable: a small perturbation $\delta\rho$ induces a pertubation δa that will exponentially grow in time. Einstein's fix did not work, the Universe *must* be dynamical.

$$H^2 = \frac{\Lambda}{3} + \frac{8\pi G}{3}(\rho_c + \delta\rho) - \frac{k}{a^2}$$

$$= \Lambda + \frac{8\pi G}{3}\delta\rho - \frac{k}{a^2}$$

$$\frac{\ddot a}{a} = \frac{\Lambda}{3} - \frac{4\pi G}{3}(\rho_c + \delta\rho)$$

$$= -\frac{4\pi G}{3}\delta\rho$$

$$\delta\dot\rho = -3\frac{\dot a}{a}(\rho_c + \delta\rho)$$

$$\approx -3\frac{\dot a}{a}\rho_c$$

Exercise 4: TOV Inequalities

a) Use the Tolman-Oppenheimer-Volkoff equations for dm/dr and $d\Phi/dr$ to eliminate ρ and p from the equation for dp/dr. Show that the resulting equation can be written as

$$\left(r^{-1}e^{-\Psi}\zeta'\right)' = \left(mr^{-3}\right)' e^{\Psi}\zeta\,, \tag{22}$$

where

$$e^{2\Psi} := \left(1-\frac{2m}{r}\right)^{-1}, \qquad \zeta := e^{\Phi}\,, \tag{23}$$

and the primes denote d/dr derivatives.

$$\frac{dm}{dr} = 4\pi r^2\rho \qquad \frac{d\Phi}{dr} = \frac{m+4\pi r^3 p}{r(r-2m)} \qquad \frac{dp}{dr} = -(p+\rho)\frac{(m+4\pi r^3p)}{r(r-2m)}$$

$$r(r-2m)\frac{d\phi}{dr} - m = 4\pi r^3 p$$

$$\frac{dp}{dr} = -\frac{1}{4\pi r^2}\left(4\pi r^2 p + 4\pi r^2\rho\right)\frac{d\phi}{dr}$$

$$= -\frac{1}{4\pi r^2}\left[\frac{dm}{dr} + \frac{1}{r}\left(r(r-2m)\frac{d\phi}{dr} - m\right)\right]\frac{d\phi}{dr}$$

But also $$\frac{d}{dr}\left[r(r-2m)\frac{d\phi}{dr} - m\right] = (r-2m)\frac{d\phi}{dr} + r(1-2m')\frac{d\phi}{dr} - m'$$

$$= 2(r-m-rm')\frac{d\phi}{dr} - m'$$

$$= \frac{d}{dr}\left[4\pi r^3 p\right]$$

$$= 12\pi r^2 p + 4\pi r^3\frac{dp}{dr}$$

$$4\pi r^3\frac{dp}{dr} = -r\left[\frac{dm}{dr} + \frac{1}{r}\left(r(r-2m)\frac{d\phi}{dr} - m\right)\right]\frac{d\phi}{dr}$$

$$= 2(r-m-rm')\frac{d\phi}{dr} - m' - 12\pi r^2 p$$

$$= 2(r-m-rm')\frac{d\phi}{dr} - m' - \frac{3}{r}\left[r(r-2m)\frac{d\phi}{dr} - m\right]$$

$$= 2(r-m-rm')\frac{d\phi}{dr} - m' - 3(r-2m)\frac{d\phi}{dr} + \frac{3m}{r}$$

$$= (-r+5m-2rm')\frac{d\phi}{dr} - m' + \frac{3m}{r}$$

Now we have

$$-r\left[\frac{dm}{dr} + \frac{1}{r}\left(r(r-2m)\frac{d\phi}{dr} - m\right)\right]\frac{d\phi}{dr} = (-r+5m-2rm')\frac{d\phi}{dr} - m' + \frac{3m}{r}$$

$$\cancel{-rm'\phi'} + (r-\cancel{2m})\phi' - \cancel{\frac{m}{r}} = (-2r + 7m - 2rm')\phi' - m' + \frac{4m}{r}$$

$$(-2r+7m-rm')\phi' - m' + \frac{4m}{r} = 0$$

a) Use the Tolman-Oppenheimer-Volkoff equations for dm/dr and $d\Phi/dr$ to eliminate ρ and p from the equation for dp/dr. Show that the resulting equation can be written as

$$\left(r^{-1}e^{-\Psi}\zeta'\right)' = \left(mr^{-3}\right)' e^{\Psi}\zeta\,, \tag{22}$$

where

$$e^{2\Psi} := \left(1 - \frac{2m}{r}\right)^{-1}, \qquad \zeta := e^{\Phi}\,, \tag{23}$$

and the primes denote d/dr derivatives.

$$(r^{-1}e^{-\Psi}\zeta')' = -\frac{1}{r^2}e^{-\Psi}\zeta' - \Psi' r^{-1}e^{-\Psi}\zeta' + r^{-1}e^{-\Psi}\zeta''$$

$$(mr^{-3})' e^{\Psi}\zeta = m' r^{-3} e^{\Psi}\zeta - 3mr^{-4}e^{\Psi}\zeta$$

$$\zeta' = \phi' e^{\phi}, \quad \zeta'' = (\phi'^2 + \phi'')e^{\phi}$$

$$-\frac{1}{r^2}\phi' - \Psi' r^{-1}\phi' + r^{-1}(\phi'^2 + \phi'') = (m' r^{-3} - 3mr^{-4})e^{2\Psi}$$

$$\left(1 - \frac{2m}{r}\right)\left[-\phi'\right.$$

b) Use $d\rho/dr \le 0$ to argue that $(mr^{-3})' \le 0$.

$$(mr^{-3})' = m' r^{-3} - 3mr^{-4}$$

Since $m' = 4\pi r^2 \rho$,

$$(mr^{-3})' = \frac{4\pi r^3 \rho - 3m}{r^4}$$

Now

$$m(r) = \int_0^r 4\pi \xi^2 \rho(\xi) d\xi$$
$$\le \int_0^r 4\pi \xi^2 \rho(0) d\xi$$
$$= \frac{4}{3}\pi r^3 \rho(0)$$

Hence

$$4\pi r^3 \rho(r) - 3m(r) \le 4\pi r^3 \rho(0) - 3m(r) = 0$$

c) It follows that $(r^{-1}e^{-\Psi}\zeta')' \le 0$. By integrating this equation from r_1 to r and using the equation for $d\Phi/dr$, show that, for any $r_1 \le r$,

$$\zeta'(r_1) \ge \left(1 - \frac{2m(r)}{r}\right)^{-1/2} \left(\frac{m(r)}{r^3} + 4\pi p(r)\right) \zeta(r)\, r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2}. \tag{24}$$

$$(r^{-1}e^{-\Psi}\zeta')' \le 0$$

$$\int_{r_1}^{r} (\xi^{-1}e^{-\Psi}\zeta')' d\xi = r^{-1}e^{-\Psi}\zeta' - r_1^{-1}e^{-\Psi(r_1)}\zeta'(r_1) \le 0$$

$$\zeta'(r_1) \ge r_1 e^{\Psi(r_1)} r^{-1} e^{-\Psi}\zeta'$$

$$e^{2\Psi} = \left(1 - \frac{2m}{r}\right)^{-1}$$

$$\zeta' = \Phi'\zeta$$

$$\frac{d\Phi}{dr} = \frac{m + 4\pi r^3 p}{r(r-2m)}$$

$$\zeta'(r_1) \ge r_1 \left(1 - \frac{2m}{r_1}\right)^{-1/2} \frac{1}{r} \zeta(r) \left(1 - \frac{2m}{r}\right)^{1/2} \frac{m + 4\pi r^3 p}{r(r-2m)}$$

$$= r_1 \left(1 - \frac{2m}{r_1}\right)^{-1/2} \frac{1}{r^3} \zeta(r) \left(1 - \frac{2m}{r}\right)^{1/2} (m + 4\pi r^3 p)\left(1 - \frac{2m}{r}\right)^{-1}$$

$$= r_1 \left(1 - \frac{2m}{r_1}\right)^{-1/2} \zeta(r) \left(1 - \frac{2m}{r}\right)^{-1/2} \left(\frac{m}{r^3} + 4\pi p\right)$$

d) Now, $(mr^{-3})' \le 0$ implies $m(r_1)/r_1 \ge m(r)r_1^2/r^3$ for $r_1 \le r$. Use this to show

$$\int_0^r r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1 \ge \frac{r^3}{2m(r)}\left[1 - \left(1 - \frac{2m(r)}{r}\right)^{1/2}\right]. \tag{25}$$

$$\frac{m(r_1)}{r_1} \ge \frac{m(r)r_1^2}{r^3}$$

$$1 - \frac{2m(r_1)}{r_1} \le 1 - \frac{2m(r)r_1^2}{r^3}$$

$$\left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} \ge \left(1 - \frac{2mr_1^2}{r^3}\right)^{-1/2}$$

$$\int_0^r r_1\left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1 \ge \int_0^r r_1\left(1 - \frac{2mr_1^2}{r^3}\right)^{-1/2} dr_1$$

Evaluate the integral on the right first with $u = r_1^2$, $du = 2r_1 dr_1$

$$\int_0^r r_1\left(1 - \frac{2mr_1^2}{r^3}\right)^{-1/2} dr_1 = \frac{1}{2}\int_0^{r^2}\left(1 - \frac{2mu}{r^3}\right)^{-1/2} du$$

$$= \frac{1}{2}\left[\left(1-\frac{2mu}{r^3}\right)^{1/2}\left(-\frac{2r^3}{2m}\right)\right]_0^{r^2}$$

$$= -\frac{r^3}{2m}\left[\left(1-\frac{2mu}{r^3}\right)^{1/2}\right]_0^{r^2}$$

$$= \frac{r^3}{2m}\left[1-\left(1-\frac{2mr^2}{r^3}\right)^{1/2}\right]$$

$$= \frac{r^3}{2m}\left[1-\left(1-\frac{2m}{r}\right)^{1/2}\right]$$

$$\int_0^r r_1\left(1-\frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1 \geq \frac{r^3}{2m}\left[1-\left(1-\frac{2m}{r}\right)^{1/2}\right]$$

e) Deduce that

$$\zeta(0) \leq \zeta(r)\left\{1-\left(\frac{1}{2}+\frac{2\pi r^3 p(r)}{m(r)}\right)\left[\left(1-\frac{2m(r)}{r}\right)^{-1/2}-1\right]\right\}. \qquad (26)$$

$$\int_0^r \zeta'(r_1)\,dr_1 \geq \left(1-\frac{2m(r)}{r}\right)^{-1/2}\left(\frac{m(r)}{r^3}+4\pi p(r)\right)\zeta(r)\int_0^r r_1\left(1-\frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1$$

$$\zeta(r)-\zeta(0) \geq \left(1-\frac{2m}{r}\right)^{-1/2}\left(\frac{m}{r^3}+4\pi p\right)\zeta(r)\frac{r^3}{2m}\left[1-\left(1-\frac{2m}{r}\right)^{1/2}\right]$$

$$\zeta(0) \leq \zeta(r) - \left(1-\frac{2m}{r}\right)^{-1/2}\left(\frac{m}{r^3}+4\pi p\right)\zeta(r)\frac{r^3}{2m}\left[1-\left(1-\frac{2m}{r}\right)^{1/2}\right]$$

$$= \zeta(r)\left\{1-\left(\frac{1}{2}+\frac{2\pi p r^3}{m}\right)\left[\left(1-\frac{2m}{r}\right)^{-1/2}-1\right]\right\}$$

f) By definition $\zeta(r)$ is positive everywhere. In particular, $\zeta(0) > 0$. Deduce that

$$\frac{m(r)}{r} \leq \frac{2}{9}\left(1-6\pi r^2 p(r)+\sqrt{1+6\pi r^2 p(r)}.\right)$$

Obviously

$$0 < \zeta(r)\left\{1-\left(\frac{1}{2}+\frac{2\pi p r^3}{m}\right)\left[\left(1-\frac{2m}{r}\right)^{-1/2}-1\right]\right\}$$

$$0 < 1-\left(\frac{1}{2}+\frac{2\pi p r^3}{m}\right)\left[\left(1-\frac{2m}{r}\right)^{-1/2}-1\right]$$

$$\left(\frac{1}{2}+\frac{2\pi p r^3}{m}\right)\left[\left(1-\frac{2m}{r}\right)^{-1/2}-1\right] < 1$$

$$\left(1-\frac{2m}{r}\right)^{-1/2} < \left(\frac{1}{2}+\frac{2\pi p r^3}{m}\right)^{-1}+1$$

$$1-\frac{2m}{r} > \left[\left(\frac{1}{2}+\frac{2\pi p r^3}{m}\right)^{-1}+1\right]^{-2}$$

Exercise 5: Solving TOV for constant density stars

A greatly simplified but instructive solution to the TOV equations is found by assuming

$$\rho(r) = \begin{cases} \rho_0 & r < R, \\ 0 & r > R, \end{cases} \tag{29}$$

where $\rho_0 > 0$ is some positive constant and R is the radius of the star. That is, we are considering a star made from incompressible matter.

a)* Integrate TOV1 to find $m(r)$. Identify the total energy of the star M.

$$\frac{dm}{dr} = 4\pi r^2 \rho(r)$$

$$m(r) = \int_0^r 4\pi \xi^2 \rho(\xi)\, d\xi$$

$$= \begin{cases} \frac{4}{3}\pi r^3 \rho_0, & r \le R \\ \frac{4}{3}\pi R^3 \rho_0, & r > R \end{cases}$$

b)* Define $x := p/\rho_0$ to write TOV3 as

$$\frac{dx}{dr} = -(1+x)(1+3x)\frac{4\pi\rho_0 r}{3 - 8\pi\rho_0 r^2}.$$

$$\frac{dp}{dr} = -(p+\rho)\frac{(m + 4\pi r^3 p)}{r(r-2m)}$$

$$\frac{dx}{dr} = \frac{1}{\rho_0}\frac{dp}{dr}$$

$$\frac{dx}{dr} = -\left(\frac{p}{\rho_0} + 1\right)\frac{m + 4\pi r^3 p}{r(r-2m)}$$

$$= -(x+1)\rho_0 \frac{\frac{4}{3}\pi r^3 + 4\pi r^3 x}{r(r-2m)}$$

$$= -\frac{4}{3}\pi r^3 (1+x)\rho_0 \frac{1+3x}{r(r-2m)}$$

$$= -\frac{4}{3}\pi r^3 (1+x)(1+3x)\rho_0 \frac{1}{r}\frac{1}{r - \frac{8}{3}\pi r^3 \rho_0}$$

$$= -(1+x)(1+3x)\frac{4\pi\rho_0 r}{3 - 8\pi\rho_0 r}$$

c)* Integrate the previous ODE while imposing $p(R) = 0$ and $p(0) = p_c$ to find

$$p_c = \rho_0 \frac{1 - \sqrt{1 - 2M/R}}{3\sqrt{1 - 2M/R} - 1}.$$

Boundary conditions mean $x(R) = 0$ and $x(0) = \frac{p_c}{\rho_0}$

$$\frac{dx}{dr} = -(1+x)(1+3x)\frac{4\pi\rho_0 r}{3 - 8\pi\rho_0 r^2}$$

$$\int_{\frac{p_c}{\rho_0}}^{0} \frac{dx}{(1+x)(1+3x)} = -\int_0^R \frac{4\pi\rho_0 r}{3 - 8\pi\rho_0 r^2}$$

```
In[2]:= Integrate[1 / ((1 + x) (1 + 3 x)), {x, 0, α}]
```

Out[2]= $\frac{1}{2}\left(-\mathrm{Log}[1+\alpha] + \mathrm{Log}[1+3\alpha]\right)$ if $\mathrm{Re}[\alpha] > 0 \,||\, -\frac{1}{3} < \mathrm{Re}[\alpha] < 0 \,||\, \alpha \notin \mathbb{R}$

```
In[5]:= Assuming[ρ0 > 0 && R > 0, Integrate[4 π ρ0 r / (3 - 8 π ρ0 r^2), {r, 0, R}]]
```

Out[5]= $-\frac{1}{4}\mathrm{Log}\left[1 - \frac{8}{3}\pi R^2 \rho_0\right]$ if $8\pi R^2 \rho_0 < 3$

(all conditions can be easily seen to be satisfied)

$$\frac{1}{2}\left[-\ln\left(1 + \frac{p_c}{\rho_0}\right) + \ln\left(1 + \frac{3p_c}{\rho_0}\right)\right] = -\frac{1}{4}\ln\left[1 - \frac{8}{3}\pi R^2 \rho_0\right]$$

$$\frac{1}{2}\ln\frac{1 + \frac{3p_c}{\rho_0}}{1 + \frac{p_c}{\rho_0}} = -\frac{1}{4}\ln\left[1 - \frac{8}{3}\pi R^2 \rho_0\right]$$

$$\left[\frac{1 + \frac{3p_c}{\rho_0}}{1 + \frac{p_c}{\rho_0}}\right]^{1/2} = \left(1 - \frac{8}{3}\pi R^2 \rho_0\right)^{-1/4}$$

Take to the 4th power & solve

$$\mathrm{Solve}\left[\frac{\left(1 + \frac{3p_c}{\rho_0}\right)^2}{\left(1 + \frac{p_c}{\rho_0}\right)^2} == \frac{1}{1 - \frac{8}{3}\pi R^2 \rho_0},\ p_c\right]$$

$$\left\{\left\{p_c \to \frac{3\rho_0 - 12\pi R^2 \rho_0^2 - \sqrt{3}\sqrt{3\rho_0^2 - 8\pi R^2 \rho_0^3}}{12\left(-1 + 3\pi R^2 \rho_0\right)}\right\}, \left\{p_c \to \frac{3\rho_0 - 12\pi R^2 \rho_0^2 + \sqrt{3}\sqrt{3\rho_0^2 - 8\pi R^2 \rho_0^3}}{12\left(-1 + 3\pi R^2 \rho_0\right)}\right\}\right\}$$

$$p_c = \frac{3\rho_0 - 12\pi R^2\rho_0^2 - \sqrt{3}\sqrt{3\rho_0^2 - 8\pi R^2\rho_0^3}}{12(-1+3\pi R^2\rho_0)}$$

Note $\frac{4}{3}\pi R^3\rho_0 = M$

$$\frac{p_c}{\rho_0} = \frac{3 - 12\pi R^2\rho_0 - \sqrt{3}\sqrt{3 - 8\pi R^2\rho_0}}{12(-1+3\pi R^2\rho_0)}$$

$$= \frac{3 - 9\frac{M}{R} - \sqrt{3}\sqrt{3-6\frac{M}{R}}}{12(-1+\frac{9}{4}\frac{M}{R})}$$

$$= \frac{1 - \frac{3M}{R} - \sqrt{1-2\frac{M}{R}}}{(-4+9\frac{M}{R})}$$

Define $s = \sqrt{1-2\frac{M}{R}}$

$$s^2 = 1 - 2\frac{M}{R}, \quad \frac{M}{R} = \frac{1-s^2}{2}$$

$$\frac{p_c}{\rho_0} = \frac{1 - \frac{3}{2}(1-s^2) - s}{-4 + \frac{9}{2}(1-s^2)}$$

$$= \frac{\frac{3}{2}s^2 - s - \frac{1}{2}}{-\frac{9}{2}s^2 + \frac{1}{2}}$$

$$= \frac{3s^2 - 2s - 1}{-9s^2 + 1}$$

$$= \frac{(1+3s)(s-1)}{(1-3s)(1+3s)}$$

$$= \frac{s-1}{1-3s} = \frac{1-s}{3s-1}$$

$$= \frac{1 - \sqrt{1-\frac{2M}{R}}}{3\sqrt{1-\frac{2M}{R}} - 1}$$

as desired

d)* Impose $p_c < \infty$ to find Buchdahl's inequality for constant density stars,

$$\frac{M}{R} < \frac{4}{9}.$$

Show that constant density stars can get arbitrarily close to saturating this bound.

The only thing that can force p to blow up is the denominator.

Thus we require

$$3\sqrt{1-\frac{2M}{R}} - 1 > 0$$

$$3\sqrt{1-\frac{2M}{R}} > 1$$

$$1 - \frac{2M}{R} > \frac{1}{9}$$

$$\frac{8}{9} > \frac{2M}{R}$$

$\frac{M}{R} < \frac{4}{9}$, which was also clear from the expression earlier $\frac{p_c}{\rho_0} = \frac{1-\frac{3M}{R} - \sqrt{1-2\frac{M}{R}}}{(-4+9\frac{M}{R})}$